

MANTUO ORIGINAL FOUNDATION EDITION - ESAT - V0.1.0

ESAT Science Modules Starting Review Notes

ESAT 理科模块起点复习笔记

A programme-aware starting review for the Physics, Chemistry and Biology modules you actually need.

按申请专业只显示你实际需要的 Physics、Chemistry 和 Biology 知识单元、课程差距与第一轮复习方法。

The logo consists of the letters 'MT' in a white, bold, sans-serif font, centered within a solid purple square.

MANTUO ADMISSION TEST LIBRARY

不再为升学考试而焦虑

Mantou original foundation edition - independently authored for ESAT preparation

How to Use This Pack

如何使用这份笔记

IN THIS EDITION	Thirty-five top-level science units, curriculum bridges, nine question methods, three original worked examples and nine active-recall prompts, filtered by the student's programme modules. Physics、Chemistry、Biology 共 35 个一级知识单元；A-Level/IAL、IB、AP 课程衔接；每模块 3 套起题方法、1 道原创逐步例题和 3 组主动回忆。页面按本人已选择的专业模块过滤。
GO DEEPER	After the knowledge map, use the ESAT Advanced Strategy Notes for module-specific playbooks, worked cases and training prescriptions, then verify Physics, Chemistry or Biology performance in timed online papers. 完成知识地图后，进入 ESAT 深度策略笔记学习模块化解题流程、完整案例和训练处方，再用在线限时试卷验证 Physics、Chemistry 或 Biology 的真实表现。
PUBLICATION BOUNDARY	This resource was independently written by Mantou. The official Content Specification and module guides are used only to verify exam facts, scope and terminology; no official questions, examples, exercises or explanations are reproduced. 本资料由满托独立撰写。官方 Content Specification 与三份模块 Guide 只用于核对考试事实、知识范围和术语；不复制官方题目、例题、练习或讲解。

FACTS FIRST

Exam Map

考试地图

PER MODULE 每个模块	27 multiple-choice questions 27 道选择题
TIME 模块时间	40 minutes, separately timed 40 分钟，独立计时
TOOLS 工具	Calculators are not permitted 不可使用计算器
SCORING 计分	One mark per correct answer; no negative marking 答对 1 分，答错不倒扣
SHARED FOUNDATION 共同基础	All modules assume Mathematics 1 knowledge 所有模块都默认 Mathematics 1 知识

A-LEVEL - IB - AP

Curriculum Bridge

课程衔接：具体缺什么

A curriculum can indicate likely exposure, not mastery. Confirm every claim against the student profile and timed evidence.

课程范围只说明可能学过什么，不等于已经掌握；请结合本人档案和真实练习。

A-Level / International A-Level Sciences	STRONG START 对应科目通常覆盖较多
LIKELY TRANSFER	- Core concepts and calculations from the corresponding A-Level Physics, Chemistry or Biology course 对应 A-Level Physics、Chemistry 或 Biology 的核心概念与计算 - Graphs, experimental variables, units, standard equations and scientific terminology 图表、实验变量、单位、标准方程和科学术语
CHECK ONE BY ONE	- Whether the student's exact exam board covers every ESAT top-level unit, including cross-topic and GCSE/IGCSE material 本人实际考试局是否包含全部 ESAT 一级单元，尤其跨章节和 GCSE/IGCSE 回溯内容 - Whether knowledge can be recalled without a calculator under a 40-minute mixed-topic multiple-choice format, not only in chapter-based homework 能否在无计算器、40 分钟和混合主题选项下快速调用，而不只是在分章节作业中会做
FIRST ACTION	Audit the units in each required module, then record unfamiliar content separately from knowledge that is understood but slow to retrieve. 先按本人需要的模块逐项核对知识单元，再把不熟的单元与限时调用问题分开记录。

IB Diploma Sciences	PARTIAL HL/SL 与选科决定覆盖
LIKELY TRANSFER	- HL courses usually provide a strong starting point in core principles, data analysis and experimental evaluation HL 通常对核心原理、数据分析和实验评价有较强起点 - Cross-topic explanations, graph trends and evidence-based reasoning usually transfer well 跨主题解释、图像趋势和证据判断通常可以直接迁移
CHECK ONE BY ONE	- Whether the student's SL/HL course and syllabus edition correspond to the exact ESAT units SL 或旧/新 syllabus 与 ESAT 的具体单位是否完全对应 - Whether UK-curriculum reactions, qualitative tests, radioactivity, plant transport or inheritance diagrams have direct curriculum evidence 英国课程常用反应、定性检验、放射性、植物运输或遗传图解是否有直接课程证据
FIRST ACTION	Do not infer readiness from the HL or SL label alone; use the 35-unit checklist to find units with partial or no curriculum evidence. 不要按 HL/SL 标签直接判定；用 35 单元清单找到没有课程证据或只部分覆盖的条目。

AP Physics / Chemistry / Biology	PARTIAL 需要按具体 AP 组合核对
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LIKELY TRANSFER	- AP Chemistry and AP Biology can cover substantial core principles and data reasoning AP Chemistry 与 AP Biology 可覆盖大量核心原理和数据推理 - A combination of AP Physics 1, 2 and C courses can cover different parts of mechanics, electromagnetism, waves and thermal physics AP Physics 1/2/C 的组合可覆盖不同部分的力学、电磁、波与热学
CHECK ONE BY ONE	- A single AP Physics course rarely demonstrates coverage of all seven Physics units 单一 AP Physics 课程通常不能证明 Physics 七个单元全部覆盖 - Whether qualitative chemistry, periodic trends, metal extraction, plant physiology and human systems were taught in the student's courses 化学定性检验、族趋势、金属提取，以及植物生理和人体系统是否在本人课程中学到
FIRST ACTION	Build evidence from the student's exact AP combination; never treat one AP subject as proof of coverage for an entire ESAT module. 先按具体 AP 科目组合建立证据，不把一门 AP 课程等同于整个 ESAT 模块。

MODULE 01 - ESAT

Physics - Models, Units and Evidence

Physics: 模型、单位与证据

Physics is not about memorising more formulae. It is about turning words, diagrams, units and conservation laws into one usable model, then rejecting impossible options with limits and orders of magnitude.

Physics 的关键不是背更多公式，而是把文字、图、单位和守恒关系快速连成同一个模型，并用边界和数量级排除不可能的选项。

LEARNING OUTCOMES	- Audit curriculum evidence and gaps across P1-P7 逐项识别 P1-P7 的课程证据与缺口 - Select the minimum sufficient equation from a diagram or written situation 从图示或文字选择最小充分方程 - Check an answer through units, direction, limiting behaviour and order of magnitude 用单位、方向、极限与数量级检查答案
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KNOWLEDGE MAP

Check Every Knowledge Unit

逐项核对知识单元

P1	Electricity 静电、电路与电功率
P2	Magnetism 磁场、电磁感应与变压器
P3	Mechanics 运动、力、动量与能量

P4	Thermal physics 传热与热容量
P5	Matter 物态、气体、密度与压强
P6	Waves 波、光、声与电磁波谱
P7	Radioactivity 原子结构、衰变与半衰期

QUESTION METHODS

What to Notice, How to Start, What to Check

看到什么，怎样开始，如何检查

01	DEFINE THE SYSTEM 先画系统边界
TRIGGER	The question contains several objects, branches, energy transfers or initial and final states. 题目同时出现多个物体、支路、能量转移或前后状态。
ACTION	Mark the system under study, known directions and initial and final states before choosing a conservation or connection rule. 先标清研究对象、已知方向和初末状态，再选择守恒或连接关系。
CHECK	Every quantity in the equation must belong to the same system and the same time interval. 方程中的每个量必须属于同一系统和同一时间区间。
02	UNITS BEFORE ALGEBRA 单位先于代数
TRIGGER	Values contain prefixes or compound units, or the requested answer requires a conversion. 数值带前缀、复合单位或题目要求换算输出。
ACTION	Write the target unit and complete necessary conversions before substitution. In a non-calculator paper, simplify factors before multiplying. 先写目标单位和必要换算，再代入最少方程；无计算器时优先约分。
CHECK	Dimensions, order of magnitude, sign and direction must all be physically sensible. 答案量纲、数量级、正负和方向必须同时合理。
03	TEST THE LIMIT 用极限排除

TRIGGER	The question asks how a changing parameter affects a graph or quantity, and options differ mainly in trend. 题目问参数增大、图像变化或四个选项只在趋势上不同。
ACTION	Imagine the variable approaching zero or becoming very large, predict the physical outcome, then compare that prediction with the options. 把变量想成接近零或很大，判断物理量应趋向什么，再比较选项。
CHECK	The predicted trend must respect conservation, steady-state behaviour and known boundaries. 趋势不能违反守恒、稳定状态或已知边界。

ORIGINAL WORKED EXAMPLE

Mantou original - total current in parallel branches

满托原创例题：并联支路的总电流

PROBLEM	A 12 V supply is connected to 6-ohm and 4-ohm resistors in parallel. What total current does the supply provide? 一个 12 V 电源连接两个并联电阻，电阻分别为 6 Ω 和 4 Ω 。忽略电源内阻，电源提供的总电流是多少？
Identify the connection 确认连接关系	Parallel branches have the same potential difference, so each resistor has 12 V across it. 并联支路两端电压相同，因此两个电阻各自承受 12 V。 The voltage across each branch is 12 volts
Find each branch current 分别求支路电流	Apply $I = V/R$ to each branch separately. 对每个支路使用 $I = V/R$ 。 The branch currents are 2 amperes and 3 amperes
Combine currents at the junction 在节点合并	The supply current is the sum of the currents entering the two branches. 电源总电流等于流入两个支路的电流之和。 The total current is 5 amperes
CONCLUSION	The total current is 5 A. 总电流为 5 A。
COMMON TRAP	Adding 6 ohms and 4 ohms assumes a series circuit. Identify the connection before deciding which quantity is common to both components. 把 6 Ω 与 4 Ω 直接相加属于串联模型；先判断连接关系，再选择电压或电流的共同量。

Close the Notes Before You Answer

合上笔记再回答

01	What should the first line of a physics calculation establish? 物理计算写第一行时，最应该先固定什么？
ANSWER	The system, directions or states, and the target unit. 研究系统、量的方向或状态，以及目标单位。
02	Why should unit checks not be left until the end? 为什么单位检查不能只放在最后？
ANSWER	Units expose incompatible quantities before algebra and reduce non-calculator conversion errors. 单位能在选方程和代数前暴露不相容的量，也能减少无计算器换算错误。
03	When is a limiting-case check faster than exact calculation? 什么时候极限判断比精确计算更快？
ANSWER	When options differ mainly by trend, sign or boundary rather than close numerical values. 当选项主要区别在趋势、符号或边界，而不是接近的数值。

Chemistry - Particles, Ratios and Evidence

Chemistry: 粒子、比例与证据链

Chemistry questions move between macroscopic observations, particle models, equation ratios and experimental evidence. Establish the reaction and its evidence before calculating moles or charge.

Chemistry 需要在宏观现象、粒子模型、方程比例与实验观察之间切换。先确定反应身份和证据，再做摩尔或电荷计算。

LEARNING OUTCOMES

- Audit curriculum evidence and gaps across C1-C17

逐项识别 C1-C17 的课程证据与缺口

- Use particles, bonding and electron transfer to explain properties and reactions

用粒子、键合和电子变解释性质与反应

- Connect equation coefficients, units and observations into a complete evidence chain

把方程系数、单位和实验观察连成完整证据链

KNOWLEDGE MAP

Check Every Knowledge Unit

逐项核对知识单元

C1	Atomic structure 原子结构与同位素
C2	The Periodic Table 元素周期表
C3	Reactions, formulae and equations 反应、化学式与方程式
C4	Quantitative chemistry 定量化学与摩尔计算
C5	Oxidation, reduction and redox 氧化、还原与氧化还原
C6	Bonding, structure and properties 化学键、结构与性质
C7	Group chemistry 族化学与趋势
C8	Separation techniques 分离技术与色谱

C9	Acids, bases and salts 酸、碱与盐
C10	Rates of reaction 反应速率与碰撞理论
C11	Energetics 能量变化与量热
C12	Electrolysis 电解与半方程式
C13	Carbon and organic chemistry 碳与有机化学
C14	Metals 金属、反应性与提取
C15	Kinetic and particle theory 粒子理论与物态
C16	Chemical tests 气体、离子与水的化学检验
C17	Air and water 空气、水与污染

QUESTION METHODS

What to Notice, How to Start, What to Check

看到什么，怎样开始，如何检查

01	IDENTIFY THE REACTION 先写反应身份
TRIGGER	The question combines substance names, observations, conditions or an incomplete equation. 题目同时给出物质名称、观察、条件或不完整方程。
ACTION	Identify the reaction type and particle or electron change before completing formulae, states and coefficients. 先判断反应类型、粒子或电子变化，再补化学式、状态和系数。
CHECK	Atoms, total charge and every stated observation must be conserved or mutually consistent. 元素、总电荷和题目观察必须全部守恒或相容。

02	USE MOLES AS THE BRIDGE 摩尔是中间层
TRIGGER	Mass, gas volume, concentration or particle number must be compared through a chemical equation. 质量、气体体积、浓度或粒子数需要通过方程比较。
ACTION	Convert the input quantity to moles, apply the equation ratio, then convert the resulting moles to the requested quantity. 输入量先变成摩尔，按方程系数转换，再变成目标量。
CHECK	Stoichiometric coefficients compare moles, not grams, cubic decimetres or concentrations directly. 系数比作用于摩尔，不直接作用于克、立方分米或浓度。

03	LINK STRUCTURE TO PROPERTY 结构解释性质
TRIGGER	The question asks about melting point, conductivity, solubility, hardness or a reaction trend. 题目问熔点、导电、溶解、硬度或反应趋势。
ACTION	State the particles, arrangement or bonding, relevant forces or mobile particles, then derive the macroscopic property. 依次写粒子种类、排列/键合、可移动粒子或作用力，再导出宏观性质。
CHECK	The explanation must name the particle or force responsible for the property, rather than repeating the observation. 解释必须指出改变性质的具体粒子或作用力，不能只重复现象。

ORIGINAL WORKED EXAMPLE

Mantou original - carbonate mass and gas volume

满托原创例题：碳酸盐与气体体积

PROBLEM	Five grams of calcium carbonate reacts completely with excess hydrochloric acid. Using 100 grams per mole for calcium carbonate and 24 cubic decimetres per mole of gas, find the carbon dioxide volume. 5.00 g 碳酸钙与过量盐酸完全反应。取 CaCO_3 的摩尔质量为 100 g mol^{-1} ，室温下 1 mol 气体体积为 24 dm^3 。生成多少 dm^3 二氧化碳？
Convert mass to moles 质量变摩尔	Divide the given mass by the molar mass. 把给定质量除以摩尔质量。 The amount of calcium carbonate is 0.0500 moles
Apply the equation ratio 使用方程比例	One mole of calcium carbonate produces one mole of carbon dioxide, so the carbon dioxide amount is also 0.0500 mol . 每 1 mol CaCO_3 生成 1 mol CO_2 ，因此二氧化碳也是 0.0500 mol 。 The amount of carbon dioxide is 0.0500 moles

Convert moles to the target volume 摩尔变目标体积	Use the molar gas volume at room temperature supplied in the question. 使用题目给定的室温摩尔气体体积。 The gas volume is 1.20 cubic decimetres
CONCLUSION	The reaction produces 1.20 dm ³ of carbon dioxide. 生成 1.20 dm ³ CO ₂ 。
COMMON TRAP	Equation coefficients compare moles. Multiplying 5.00 g directly by 24 skips both the unit conversion and the reaction ratio. 方程系数比连接的是摩尔；把 5.00 g 直接乘 24 会跳过单位和反应比例。

ACTIVE RECALL

Close the Notes Before You Answer

合上笔记再回答

01	What is the general three-step bridge in quantitative chemistry? 定量化学的通用三步桥是什么？
ANSWER	Convert the input to moles, apply the equation ratio, then convert moles to the target quantity. 输入量变摩尔，按方程系数转换，摩尔再变成目标量。
02	Why is naming the bond alone insufficient when explaining a property? 解释物质性质时，为什么不能只写键的名称？
ANSWER	You must connect particle arrangement, forces or mobile charge carriers to the observed property. 还要说明粒子排列、作用力或可移动带电粒子怎样导致观察到的性质。
03	Which two conservation checks should follow balancing? 配平后最少要检查哪两种守恒？
ANSWER	The number of atoms of each element and the total charge. 每种元素的原子数与总电荷。

Biology - Levels, Mechanisms and Data

Biology: 层级、机制与数据

Biology questions often combine cellular mechanisms, organ systems, inheritance probabilities and experimental data. Identify the biological level first, then trace the shortest valid causal chain.

Biology 题常把细胞机制、器官系统、遗传概率和实验数据放在同一道题中。先确定讨论层级，再沿因果链解释变化。

LEARNING OUTCOMES

- Audit curriculum evidence and gaps across B1-B11

逐项识别 B1-B11 的课程证据与缺口

- Distinguish cellular, tissue, organ, organism and ecosystem levels

区分细胞、组织、器官、个体和生态层级

- Use controls, variables, proportions and causal mechanisms to interpret data

用对照、变量、比例和因果机制解释数据

KNOWLEDGE MAP

Check Every Knowledge Unit

逐项核对知识单元

B1	Cells 细胞结构与组织层次
B2	Movement across membranes 跨膜运输
B3	Cell division and sex determination 细胞分裂、繁殖与性别决定
B4	Inheritance 遗传、杂交与家系图
B5	DNA DNA、基因与蛋白质
B6	Gene technologies 基因技术
B7	Variation 变异、选择与进化
B8	Enzymes 酶与反应速率

B9	Animal physiology 动物生理与健康
B10	Ecosystems 生态系统、循环与生物多样性
B11	Plant physiology 光合作用与植物运输

QUESTION METHODS

What to Notice, How to Start, What to Check

看到什么，怎样开始，如何检查

01	IDENTIFY THE LEVEL 先确定生物层级
TRIGGER	The question combines molecular, cellular, organ, organism or population descriptions. 题目同时出现分子、细胞、器官、个体或种群表述。
ACTION	Circle the level at which the direct change occurs, then state how it affects the level immediately above or below it. 圈出直接变化发生的层级，再写它怎样影响上一级或下一级。
CHECK	Do not use an organism's purpose to explain a molecular mechanism, and do not turn correlation into causation. 不要用个体目的解释分子机制，也不要把相关性直接写成因果。
02	BUILD THE CAUSAL CHAIN 沿机制写因果链
TRIGGER	The question asks why a change produces a phenotype, rate or system-level result. 题目问某变化为什么导致某表型、速率或系统结果。
ACTION	Write the shortest chain from stimulus or change, through the direct mechanism and intermediate result, to the observable outcome. 按刺激/变化 => 直接机制 => 中间结果 => 可观察结果写最短链。
CHECK	Every arrow in the chain must be supported by a taught process or structural relationship. 每个箭头都必须能用课程中的过程或结构解释。
03	CHECK THE CONTROL 实验先查对照
TRIGGER	The question presents a table, graph, repeat measurements or asks for an evaluation of a conclusion. 题目给表格、图像、重复测量或要求评价结论。

ACTION	Identify the independent variable, dependent variable, controlled variables, control and repeats before comparing proportions, trends and anomalies. 先找自变量、因变量、控制变量、对照和重复，再比较比例、趋势与异常。
CHECK	The conclusion must not extend beyond the treatment, sample and variables that were actually measured. 结论范围不能超过实验处理、样本和实际测量变量。

ORIGINAL WORKED EXAMPLE

Mantou original - probability of a recessive phenotype

满托原创例题：隐性性状的后代概率

PROBLEM	An autosomal recessive trait is caused by allele a. One parent is Aa and the other is aa. What is the probability that one child shows the trait? 某常染色体隐性性状由等位基因 a 引起。一位亲本基因型为 Aa，另一位为 aa。一个子代表现该隐性性状的概率是多少？
List the gametes 先写配子	The Aa parent produces A and a gametes with equal probability; the aa parent can produce only a gametes. Aa 亲本产生 A 或 a 配子，概率各一半；aa 亲本只能产生 a 配子。 The first parent produces A or a equally; the second produces only a
Combine offspring genotypes 组合后代基因型	The possible offspring are Aa and aa, each with probability one half. 可能后代为 Aa 或 aa，概率各一半。 The probabilities of Aa and aa are each one half
Map genotype to phenotype 从基因型到表型	The recessive phenotype appears only in an aa individual. 隐性性状只在 aa 时表现。 The recessive phenotype probability is one half
CONCLUSION	The probability is 1/2, or 50%. 概率为 1/2，即 50%。
COMMON TRAP	An aa parent does not make every child recessive. The allele contributed by the other parent still determines half of the combinations. 不能因为一个亲本是 aa 就判定所有后代都表现隐性；另一个亲本提供的等位基因仍决定一半组合。

Close the Notes Before You Answer

合上笔记再回答

01	What is the first step in a biological causal question? 看到一个生物学因果题，第一步是什么？
ANSWER	Identify the level where the direct change occurs and the starting and ending states to connect. 确定直接变化发生在哪个层级，并列需要连接的起点和结果。
02	Why does a trend in experimental data not automatically prove causation? 实验图像有趋势时，为什么还不能立刻写因果？
ANSWER	Controls, variables, sample, repeats and alternative explanations must support the causal claim. 需要检查对照、控制变量、样本、重复和替代解释是否足以支持因果范围。
03	Why separate genotype from phenotype in inheritance questions? 遗传题中基因型与表型为什么要分两步？
ANSWER	Gamete combinations give genotype probabilities; expression rules then map genotypes to phenotypes. 配子组合先给出基因型概率，显隐性或其他表达规则再把基因型映射为表型。

Review Loop

把复习变成下一步行动

01	FILTER BY PROGRAMME 按专业锁定模块	Study only the Physics, Chemistry or Biology modules required by the chosen programme, rather than copying another student's combination. 只复习申请专业实际要求的 Physics、Chemistry 或 Biology，不按别人组合浪费时间。
02	MAP COURSE EVIDENCE 按课程核对证据	Mark each top-level unit as covered, partially covered or unsupported by the student's course. A course label is not evidence of mastery. 把每个一级单元标为课程已覆盖、部分覆盖或没有证据；课程标签本身不等于掌握。
03	RECALL THEN PRACTISE 主动回忆后练习	Close the notes and answer the recall prompts before taking a short module diagnostic. Record whether each miss came from content, model selection or timed retrieval. 合上笔记回答主动回忆，再进入对应模块短诊断；记录是知识缺口、模型选择还是限时调用问题。
04	UPDATE FROM EVIDENCE 用作答证据更新	Update the next review cycle from actual answers, time and answer changes; never infer a score or admission probability directly from curriculum background. 根据真实作答、用时和改答更新下一轮计划，不从课程背景直接推断分数或录取概率。

Version Boundary and Official Anchors

版本边界与官方依据

This resource was independently written by Mantou. The official Content Specification and module guides are used only to verify exam facts, scope and terminology; no official questions, examples, exercises or explanations are reproduced.

本资料由满托独立撰写。官方 Content Specification 与三份模块 Guide 只用于核对考试事实、知识范围和术语；不复制官方题目、例题、练习或讲解。

After the knowledge map, use the ESAT Advanced Strategy Notes for module-specific playbooks, worked cases and training prescriptions, then verify Physics, Chemistry or Biology performance in timed online papers.

完成知识地图后，进入 ESAT 深度策略笔记学习模块化解题流程、完整案例和训练处方，再用在线限时试卷验证 Physics、Chemistry 或 Biology 的真实表现。

ESAT Content Specification - October 2026 and January 2027

Used to verify the five-module structure, 27 questions in 40 minutes, separate timing, non-calculator rule, scoring and the top-level science scope.

核对五模块结构、27 题/40 分钟、独立计时、无计算器、计分规则和理科模块一级知识范围。

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ESAT Guide - Physics

Used only to verify the P1-P7 Physics scope and official terminology; all methods, examples and curriculum bridges here are independently written.

只核对 Physics 的 P1-P7 范围与官方术语；本站方法、例题和课程桥接均为独立创作。

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ESAT Guide - Chemistry

Used only to verify the C1-C17 Chemistry scope and official terminology; no official examples or explanations are used.

只核对 Chemistry 的 C1-C17 范围与官方术语；不使用官方例题或讲解。

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ESAT Guide - Biology

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